

Using a tidy approach in R to analyze Mexico's 2024 ENVIPE

Part 1: Estimating crime prevalence with the {srvyr} package

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Overview

In this article, I will guide you through the process of analyzing Mexico's 2024 National Survey of Victimization and Perception of Public Safety (*Encuesta Nacional de Victimización y Percepción sobre Seguridad Pública*, ENVIPE) using the `{tidyverse}` framework for survey analysis with the `{srvyr}` package. Specifically, we will calculate the **crime prevalence rate per 100,000 inhabitants by state, according to the victim's sex** (*Tasa de prevalencia delictiva por entidad federativa por cada cien mil habitantes, según sexo de la víctima*).

Official tables with this indicator can be found [here](#).

By the end of this article, you will be able to use the `{tidyverse}` to:

- Load the survey microdata from INEGI's website.
- Specify the ENVIPE survey design.
- Calculate crime prevalence indicators.

About the ENVIPE

ENVIPE is an official survey that provides nationally and state-representative information on household crime victimization and perceptions of public safety in Mexico. Conducted annually since 2011 by the [National Institute of Statistics and Geography \(INEGI\)](#), ENVIPE collects data on the following crimes:

- Total vehicle theft (car, pickup truck, or truck).
- Theft of vehicle accessories, spare parts, or tools (car, pickup truck, or truck).
- Wall graffiti, intentional vehicle damage, and other types of vandalism.
- Burglary or attempted burglary.

- Kidnapping.
- Enforced disappearance.
- Homicide.
- Robbery or assault on the street or public transportation.
- Theft in a manner different from the above.
- Bank fraud.
- Consumer fraud.
- Extortion.
- Threats.
- Injuries.
- Kidnapping.
- Harassment or intimidation.
- Groping.
- Exhibitionism.
- Attempted rape.
- Sexual rape.
- Other crimes.

All survey-related information, including questionnaires, methodology, glossaries, summaries, and microdata, is fully available at [INEGI's website](#).

Set-up

First, install and load the required packages:

```
library(tidyverse) # Tidy data analysis framework
library(srvyr) # Tidy syntax for survey data analysis
library(archive) # Extract files from a ZIP archive
library(kableExtra) # Create HTML table
```

Loading survey microdata from INEGI

Microdata is available [here](#) in various formats (CSV, DBF, DTA, SAV, and RData). In this post, we will use the RData files.

```
load(archive_read(
  archive = "https://www.inegi.org.mx/contenidos/programas/envipe/2024/microdatos/bd_envipe_2
  file = "BD_ENVIPE_2024.RData"
))
```

 Tip

Using the `{archive}` package, we can directly extract the required file from `bd_envipe_2024_RData.zip`.

This loads the following objects into the global environment:

- **THogar**: This object contains the interview result, the row of the selected respondent, the household factor, and other household characteristics.
- **TMod_Vic**: This object includes information on the crimes suffered by the selected respondent and their household during the reference year, specifically during 2023. Information comes from the victimization module of the survey.
- **TPer_Vic1**: This object includes information related to the perception of public security within the respondent's geographic area, antisocial behaviors in their immediate surroundings, and changes in habits due to fear of becoming a crime victim in 2023. Additionally, it presents information regarding the performance of authorities and trust in them. In other words, this table covers Sections IV and V of the main questionnaire.
- **TPer_Vic2**: This object consolidates information related to household and personal victimization, particularly focusing on high-impact crimes experienced by the household (such as kidnapping, enforced disappearance, or homicide) or by the selected respondent. This table includes Sections VI and VII of the main questionnaire.
- **TSDem**: This object contains the socio-demographic characteristics of household members, corresponding to Section III of the main questionnaire.
- **TSVivienda**: This object contains the characteristics of the households captured through the questionnaire cover page, as well as the count of household residents and the number of households. In other words, it includes Sections I and II of the main questionnaire.

Calculating crime prevalence rate per 100,000 thousand inhabitants by state, according to the victim's sex

To estimate the crime prevalence rate, we use information from **TPer_Vic2** (screening questions on victimization from the main questionnaire) and **TMod_Vic** (specific victimization module applied to the respondents who reported being a victim of at least one of the crimes included in the survey during the reference period). **TPer_Vic2** will serve to characterize the total population and **TMod_Vic** to describe the victims and the associated crimes.

1. Specifying the ENVIPE survey design

ENVIPE follows a probabilistic, three-stage, stratified, and clustered sampling design. The sampling unit is the household, while the unit of analysis is the population aged 18 and older in selected households. More details on the sampling design can be found [here](#) (in Spanish).

From the documentation on [data structure](#) (in Spanish), the key variables for specifying the survey design are:

- ID_PER: Person identifier
- EST_DIS: Stratum
- UPM_DIS: Primary sampling unit
- FAC_ELE: Weighting factor for public security perception and victimization estimates.

We specify the survey design as follows:

```
# Identify victims from TMod_Vic, excluding
# "Wall graffiti, intentional vehicle damage, and other vandalism" (BPCOD != "03")
id_victims <- TMod_Vic |>
  filter(BPCOD != "03") |>
  pull(ID_PER)

# Create victimization variable in TPer_Vic2
TPer_Vic2 <- TPer_Vic2 |>
  mutate(victimization = ifelse(ID_PER %in% id_victims, 1, 0))

# Convert FAC_ELE (weights) to numeric
TPer_Vic2 <- TPer_Vic2 |>
  mutate(FAC_ELE = as.numeric(FAC_ELE))

# Label SEX0 variable
TPer_Vic2 <- TPer_Vic2 |>
  mutate(SEX0 = recode(SEX0, "1" = "Man", "2" = "Woman"))

# Create survey object
envipe_design <- TPer_Vic2 |>
  as_survey_design(
    weights = FAC_ELE, # Weights (inverse probability)
    strata = EST_DIS, # Stratum
    ids = UPM_DIS, # Primary sampling unit
    nest = TRUE # Nesting within strata
  )
```

2. Calculating point estimates and variability indicators

To compute proportions from survey data, we use `survey_mean()` from `{srvyr}` package. From ENVIPE's documentation, we know that `NOM_ENT` represents the state, while `SEX0` corresponds to the respondent's sex.

```
envipe_design |>
  # `interact` function is used to calculate the proportion
  # over the interaction of state and sex
  group_by(interact(NOM_ENT, SEX0)) |>
  summarize(
    Proportion = survey_mean(victimization,
      level = 0.90, # Confidence level
      vartype = c("se", "ci", "cv") # Variability indicators
    )
  ) |>
  ungroup() |>
  mutate(
    `Prevalence rate` = Proportion * 100000, # Rate per 100,000 people
    `Standard error` = Proportion_se * 100000,
    `Low 90% CI` = Proportion_low * 100000,
    `Upper 90% CI` = Proportion_upp * 100000,
    `Coefficient of variation %` = Proportion_cv * 100,
    State = str_to_sentence(NOM_ENT) # Capitalize only the first letter
  ) |>
  rename(Sex = SEX0) |>
  select(
    State, Sex, `Prevalence rate`, `Standard error`,
    `Low 90% CI`, `Upper 90% CI`, `Coefficient of variation %`
  ) |>
  kbl() |> # HTML table
  kable_styling() # Apply bootstrap theme to the table
```

The output table looks like this:

State	Sex	Prevalence rate	Standard error	Low 90% CI	Upper 90% CI
Aguascalientes	Man	33710.81	2172.7372	30136.71	37284.9
Aguascalientes	Woman	32041.57	1831.7081	29028.45	35054.6
Baja California	Man	23043.35	1604.9769	20403.20	25683.5
Baja California	Woman	24480.64	1536.1771	21953.66	27007.6
Baja California Sur	Man	19329.94	1244.1353	17283.36	21376.5
Baja California Sur	Woman	23817.58	1466.2963	21405.56	26229.6
Campeche	Man	21832.87	1455.8508	19438.03	24227.7
Campeche	Woman	23213.74	1381.0933	20941.88	25485.6
Chiapas	Man	16602.99	986.0052	14981.03	18224.9
Chiapas	Woman	12074.36	978.7862	10464.28	13684.4
Chihuahua	Man	22303.22	1234.5065	20272.49	24333.9
Chihuahua	Woman	22804.31	1181.0327	20861.54	24747.0
Ciudad De Mexico	Man	34562.66	1089.2714	32770.84	36354.4
Ciudad De Mexico	Woman	30746.76	990.3528	29117.66	32375.8
Coahuila De Zaragoza	Man	18179.70	1249.1422	16124.89	20234.5
Coahuila De Zaragoza	Woman	20447.19	1238.7111	18409.54	22484.8
Colima	Man	21106.85	1593.5970	18485.43	23728.2
Colima	Woman	21295.28	1351.1455	19072.68	23517.8
Durango	Man	15240.74	1133.1868	13376.67	17104.8
Durango	Woman	17401.18	1166.0910	15482.99	19319.3
Guanajuato	Man	20057.38	1428.0981	17708.20	22406.5
Guanajuato	Woman	18930.15	1287.0811	16812.93	21047.3
Guerrero	Man	16981.89	1494.6554	14523.22	19440.5
Guerrero	Woman	14572.44	1238.6234	12534.93	16609.9
Hidalgo	Man	22397.82	1338.0474	20196.77	24598.8
Hidalgo	Woman	20215.02	1157.0294	18311.73	22118.3
Jalisco	Man	24390.13	1548.7916	21842.40	26937.8
Jalisco	Woman	25141.90	1438.5170	22775.58	27508.2
Mexico	Man	32906.59	1632.3183	30221.47	35591.7
Mexico	Woman	33023.78	1483.3471	30583.71	35463.8
Michoacan De Ocampo	Man	14405.17	1031.1125	12709.01	16101.3
Michoacan De Ocampo	Woman	15485.20	1036.3320	13780.46	17189.9
Morelos	Man	26995.43	1517.6054	24499.00	29491.8
Morelos	Woman	24869.74	1688.5969	22092.04	27647.4
Nayarit	Man	16075.56	1516.6195	13580.76	18570.3
Nayarit	Woman	18639.61	1597.1842	16012.28	21266.9
Nuevo Leon	Man	23602.12	1261.9498	21526.25	25678.0
Nuevo Leon	Woman	21440.72	1211.6372	19447.60	23433.8
Oaxaca	Man	12940.56	1158.2366	11035.29	14845.8
Oaxaca	Woman	13548.61	1017.7020	11874.51	15222.7
Puebla	Man	26519.64	1342.0473	24312.00	28727.2
Puebla	Woman	24176.45	1124.4656	22326.73	26026.1
Queretaro	Man	30297.50	1620.6534	27631.57	32963.4
Queretaro	Woman	25746.31	1260.3524	23673.06	27819.5
Quintana Roo	Man	22592.04	1384.4250	20314.70	24869.3
Quintana Roo	Woman	21775.64	1237.8095	19739.48	23811.8
San Luis Potosi	Man	24019.02	1874.7438	20935.11	27102.9
San Luis Potosi	Woman	23227.97	1491.3111	20774.80	25681.1
Sinaloa	Man	21181.44	1122.0160	19335.76	23027.1
Sinaloa	Woman	20562.22	1086.2790	18775.32	22349.1
Sonora	Man	26864.66	1816.2677	23876.94	29852.3